

Cybersecurity Intern — Google

Overview

As a Cybersecurity Intern at Google, you will work with security engineers and researchers to help protect Google's products, infrastructure, and users from evolving security threats. You will gain hands-on experience in areas such as vulnerability analysis, threat detection, security automation, incident response, and secure software development.

You will collaborate with cross-functional teams to identify security risks, build tools to improve security processes, and develop solutions that protect systems at Google's global scale.

Responsibilities

- Assist security engineers in identifying, analyzing, and mitigating security vulnerabilities across applications, networks, and infrastructure.
 - Develop scripts and automation tools to improve security workflows, monitoring, and threat detection processes.
 - Analyze security logs, alerts, and system data to identify suspicious activities and potential security incidents.
 - Support vulnerability assessments, penetration testing, and security reviews of software systems.
 - Research emerging cybersecurity threats, attack techniques, and defensive strategies.
 - Participate in incident response activities, including investigation, documentation, and remediation.
 - Collaborate with software engineers to implement secure coding practices and improve application security.
 - Build dashboards, tools, and reports to improve visibility into security risks.
 - Document security findings and communicate technical recommendations to engineering teams.
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Minimum Qualifications

- Currently pursuing a Bachelor's or Master's degree in Computer Science, Computer Engineering, Cybersecurity, or related technical field.
 - Experience programming in one or more languages:
 - Python
 - Java
 - C++
 - Go
 - Understanding of computer science fundamentals:
 - Data structures
 - Algorithms
 - Operating systems
 - Computer networks
 - Familiarity with cybersecurity concepts:
 - Authentication and authorization
 - Encryption
 - Network security
 - Vulnerability management
 - Ability to analyze problems and communicate technical solutions.
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Preferred Qualifications

- Experience with security tools:
 - Wireshark
 - Burp Suite
 - Metasploit
 - Nmap
- Knowledge of cloud security concepts:
 - Identity and Access Management (IAM)
 - Containers
 - Virtual machines
- Experience with:
 - Linux environments
 - Git
 - Security automation
 - Machine learning applications in cybersecurity
- Previous cybersecurity projects, research, or Capture The Flag (CTF) experience.